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Selective Search for Object Recognition

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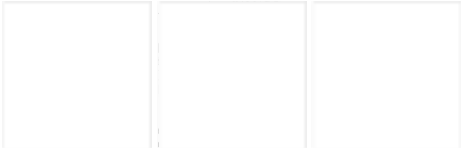
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Abstract and Figures

This paper addresses the problem of generating possible object locations for use in object recognition. We introduce selective search which combines the strength of both an exhaustive search and segmentation. Like segmentation, we use the image structure to guide our sampling process. Like exhaustive search, we aim to capture all possible object locations. Instead of a single technique to generate possible object locations, we diversify our search and use a variety of complementary image partitionings to deal with as many image conditions as possible. Our selective search results in a small set of data-driven, class-independent, high quality locations, yielding 99 % recall and a Mean Average Best Overlap of 0.879 at 10,097 locations. The reduced number of locations compared to an exhaustive search enables the use of stronger machine learning techniques and stronger appearance models for object recognition. In this paper we show that our selective search enables the use of the powerful Bag-of-Words model for recognition. The selective search software is made publicly available (Software: <http://disi.unitn.it/~uijlings/SelectiveSearch.html>).



Two examples of our selective search method... Our selective search method... The Average Best Overlap scores...

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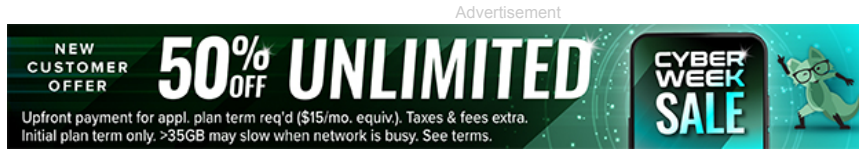
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Selective Search for Object Recognition

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Abstract

This paper addresses the problem of generating possible object locations for use in object recognition. We introduce Selective Search which combines the strength of both an exhaustive search and segmentation. Like segmentation, we use the image structure to guide our sampling process. Like exhaustive search, we aim to capture all possible object locations. Instead of a single technique to generate possible object locations, we *diversify* our search and use a variety of complementary image partitionings to deal with as many image conditions as possible. Our Selective Search results in a small set of data-driven, class-independent, high quality locations, yielding 99% recall and a Mean Average Best Overlap of 0.879 at 10,097 locations. The reduced number of locations compared to an exhaustive search enables the use of stronger machine learning techniques and stronger appearance models for object recognition. In this paper we show that our selective search enables the use of the powerful Bag-of-Words model for recognition. The Selective Search software is made publicly available ¹.

1 Introduction

For a long time, objects were sought to be delineated before their identification. This gave rise to segmentation, which aims for a unique partitioning of the image through a generic algorithm, where there is one part for all object silhouettes in the image. Research on this topic has yielded tremendous progress over the past years [3, 6, 13, 26]. But images are intrinsically hierarchical: In

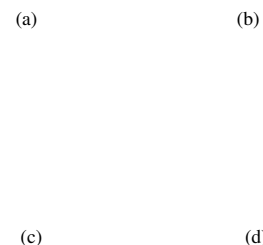


Figure 1: There is a high variety of reasons that an image forms an object. In (b) the cats can be distinguished by color texture. In (c) the chameleon can be distinguished from rounding leaves by texture, not colour. In (d) the wheels of the car because they are enclosed, not because they are in texture or colour. Therefore, to find objects in a structure it is necessary to use a variety of diverse strategies. Furthermore, an image is intrinsically hierarchical as there is no single feature which the complete table, salad bowl, and salad spoon can be distinguished by.

Figure 1a the salad and spoons are inside the salad bowl, which in turn stands on the table. Furthermore, depending on the context the term *table* in this picture can refer to only the wood or include everything on the table. Therefore both the nature of images and the different uses of an object category are hierarchical. This prohibits the unique partitioning of objects for all but the most specific purposes. Hence for most tasks multiple scales in a segmentation are a necessity. This is most naturally addressed by using a hierarchical partitioning, as done for example by Arbelaez *et al.* [3].

Besides that a segmentation should be hierarchical, a generic solution for segmentation using a single strategy may not exist at all. There are many conflicting reasons why a region should be grouped together: In Figure 1b the cats can be separated using colour, but their texture is the same. Conversely, in Figure 1c the chameleon

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¹<http://disi.unitn.it/~uijlings/SelectiveSearch.html>

is similar to its surrounding leaves in terms of colour, yet texture differs. Finally, in Figure 1d, the wheels are wildly different from the car in terms of both colour and texture, yet are by the car. Individual visual features therefore cannot remove ambiguity of segmentation.

And, finally, there is a more fundamental problem. Regions with very different characteristics, such as a face over a sweater, can only be combined into one object after it has been established that the object at hand is a human. Hence without prior recognition it is hard to decide that a face and a sweater are part of one object.

This has led to the opposite of the traditional approach: object localisation through the identification of an object. This approach in object recognition has made enormous progress in the last decade [8, 12, 16, 35]. With an appearance model from examples, an exhaustive search is performed where each location within the image is examined as to not miss any object location [8, 12, 16, 35].

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However, the exhaustive search itself has several drawbacks. Searching every possible location is computationally infeasible. The search space has to be reduced by using a regular grid, fixed scales, and fixed aspect ratios. In most cases the number of locations to visit remains huge, so much that alternative restrictions need to be imposed. The classifier is simplified and the appearance model needs to be fast. Furthermore, a uniform sampling yields many boxes for which it is immediately clear that they are not supportive of an object. Rather than sampling locations blindly using an exhaustive search, a key question is: Can we steer the sampling by a data-driven analysis?

In this paper, we aim to combine the best of the intuitions of segmentation and exhaustive search and propose a data-driven *selective search*. Inspired by bottom-up segmentation, we aim to exploit the structure of the image to generate object locations. Inspired by exhaustive search, we aim to capture all possible object locations. Therefore, instead of using a single sampling technique, we aim to *diversify* the sampling techniques to account for as many image conditions as possible. Specifically, we use a data-driven grouping-based strategy where we increase diversity by using a variety of complementary grouping criteria and a variety of complementary colour spaces with different invariance properties. The set of locations is obtained by combining the locations of these complementary partitionings. Our goal is to generate a class-independent, data-driven, selective search strategy that generates a small set of high-quality object locations.

Our application domain of selective search is object recognition. We therefore evaluate on the most commonly used dataset for this purpose, the Pascal VOC detection challenge which consists of 20 object classes. The size of this dataset yields computational constraints for our selective search. Furthermore, the use of this dataset means that the quality of locations is mainly evaluated in terms of bounding boxes. However, our selective search applies to regions as well and is also applicable to concepts such as “grass”.

In this paper we propose selective search for object recognition. Our main research questions are: (1) What are good diversification strategies for adapting segmentation as a selective search strategy? (2) How effective is selective search in creating a small set of high-

quality object locations such as HOG [8, 16, 36]. This method is often used as a building step in a cascade of classifiers [16, 36].

Related to the sliding window technique is the highly scalable part-based object localisation method of Felzenszwalb *et al.* Their method also performs an exhaustive search using SVM and HOG features. However, they search for object parts, whose combination results in an impressive detection performance.

Lampert *et al.* [17] proposed using the appearance model to guide the search. This both alleviates the constraints of regular grid, fixed scales, and fixed aspect ratio, while at the same time reduces the number of locations visited. This is done by directly searching for the optimal window within the image using a branch and bound technique. While they obtain impressive results for linear classifiers, [1] found that for non-linear class methods in practice still visits over a 100,000 windows per image.

Instead of a blind exhaustive search or a branch and bound search, we propose selective search. We use the underlying image structure to generate object locations. In contrast to the previous methods, this yields a completely class-independent set of locations. Furthermore, because we do not use a fixed aspect ratio, our method is not limited to objects but should be able to find stuff like “grass” and “sand” as well (this also holds for faces). Finally, we hope to generate fewer locations, which should make the problem easier as the variability of samples becomes lower. More importantly, it frees up computational power which can be used for stronger machine learning techniques and more complex appearance models.

2.2 Segmentation

Both Carreira and Sminchisescu [4] and Endres and Hoiem use pose to generate a set of class independent object hypotheses for segmentation. Both methods generate multiple foreground segmentations, learn to predict the likelihood that a ground segment is a complete object, and use this to rank the hypotheses. Both algorithms show a promising ability to accurately delineate objects within images, confirmed by [19] who

quality locations within an image? (3) Can we use selective search to employ more powerful classifiers and appearance models for object recognition?

2 Related Work

state-of-the-art results on pixel-wise image classification. As common in segmentation, both methods rely on a single algorithm for identifying good regions. They obtain a variety of locations by using many randomly initialised foreground and ground seeds. In contrast, we explicitly deal with a variety of conditions by using different grouping criteria and different representations. This means a lower computational investment

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As an object can be located at any position and scale in the image, it is natural to search everywhere [8, 16, 36]. However, the visual search space is huge, making an exhaustive search computationally expensive. This imposes constraints on the evaluation cost per location and/or the number of locations considered. Hence most of these sliding window techniques use a coarse search grid and fixed aspect ratios, using weak classifiers and economic image features

recognizing objects based on their parts. They first generate a set of part hypotheses using a grouping method based on *Ar et al.* [3]. Each part hypothesis is described by both appearance and shape features. Then, an object is recognized and categorized by using its parts, achieving good results for shape recognition. In their work, the segmentation is hierarchical and yields results at all scales. However, they use a single grouping

Figure 2: Two examples of our selective search showing the necessity of different scales. On the left we find many objects at different scales. On the right we necessarily find the objects at different scales as the girl is contained by the tv.

whose power of discovering parts or objects is left unevaluated. In this work, we use multiple complementary strategies to deal with as many image conditions as possible. We include the locations generated using [3] in our evaluation.

2.3 Other Sampling Strategies

Alexe *et al.* [2] address the problem of the large sampling space of an exhaustive search by proposing to search for *any* object, independent of its class. In their method they train a classifier on the

3 Selective Search

In this section we detail our selective search algorithm for object recognition and present a variety of diversification strategies with as many image conditions as possible. A selective search algorithm is subject to the following design considerations:

Capture All Scales. Objects can occur at any scale within an image. Furthermore, some objects have less clear boundaries than other objects. Therefore, in selective search a variety of scales have to be taken into account, as illustrated in Figure 2. This is most naturally achieved by using an hierarchical search algorithm.

object windows of those objects which have a well-defined shape (as opposed to stuff like “grass” and “sand”). Then instead of a full exhaustive search they randomly sample boxes to which they apply their classifier. The boxes with the highest “objectness” measure serve as a set of object hypotheses. This set is then used to greatly reduce the number of windows evaluated by class-specific object detectors. We compare our method with their work.

Another strategy is to use visual words of the Bag-of-Words model to predict the object location. Vedaldi *et al.* [34] use jumping windows [5], in which the relation between individual visual words and the object location is learned to predict the object location in new images. Maji and Malik [23] combine multiple of these relations to predict the object location using a Hough-transform, after which they randomly sample windows close to the Hough maximum. In contrast to learning, we use the image structure to sample a set of class-independent object hypotheses.

To summarize, our novelty is as follows. Instead of an exhaustive search [8, 12, 16, 36] we use segmentation as selective search yielding a small set of class independent object locations. In contrast to the segmentation of [4, 9], instead of focusing on the best segmentation algorithm [3] we use a variety of strategies to deal with as many image conditions as possible, thereby severely reducing computational costs while potentially capturing more objects accurately. Instead of learning an objectness measure on randomly sampled boxes [2], we use a bottom-up grouping procedure to generate good object locations.

Diversification. There is no single optimal strategy to group regions together. As observed earlier in Figure 1, regions form an object because of only colour, only texture, or parts are enclosed. Furthermore, lighting conditions: shading and the colour of the light may influence how regions form an object. Therefore instead of a single strategy which works well in most cases, we want to have a diverse set of strategies to deal with all cases.

Fast to Compute. The goal of selective search is to yield a set of possible object locations for use in a practical object detection framework. The creation of this set should not be a computational bottleneck, hence our algorithm should be reasonably fast.

3.1 Selective Search by Hierarchical Grouping

We take a hierarchical grouping algorithm to form the basis of selective search. Bottom-up grouping is a popular approach to segmentation [6, 13], hence we adapt it for selective search. In the process of grouping itself is hierarchical, we can naturally generate locations at all scales by continuing the grouping process until the whole image becomes a single region. This satisfies the requirement of capturing all scales.

As regions can yield richer information than pixels, we use region-based features whenever possible. To get a set of starting regions which ideally do not span multiple object

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the fast method of Felzenszwalb and Huttenlocher [13], which [3] found well-suited for such purpose.

Our grouping procedure now works as follows. We first use [13] to create initial regions. Then we use a greedy algorithm to iteratively group regions together: First the similarities between all neighbouring regions are calculated. The two most similar regions are grouped together, and new similarities are calculated between the resulting region and its neighbours. The process of grouping the most similar regions is repeated until the whole image becomes a single region. The general method is detailed in Algorithm 1.

Algorithm 1: Hierarchical Grouping Algorithm

Input: (colour) image

Output: Set of object location hypotheses L

Obtain initial regions $R = \{r_1, \dots, r_n\}$ using [13]

Initialise similarity set $S = \emptyset$

foreach Neighbouring region pair (r_i, r_j) **do**

 Calculate similarity $s(r_i, r_j)$

$S = S \cup s(r_i, r_j)$

while $S \neq \emptyset$ **do**

 Get highest similarity $s(r_i, r_j) = \max(S)$

 Merge corresponding regions $r_t = r_i \cup r_j$

 Remove similarities regarding r_i : $S = S \setminus s(r_i, r_*)$

 Remove similarities regarding r_j : $S = S \setminus s(r_*, r_j)$

 Calculate similarity set S_t between r_t and its neighbours

$S = S \cup S_t$

$R = R \cup r_t$

colour channels	R	G	B	I	V	L	a	b	S	r	g
Light Intensity	-	-	-	-	-	-	+/-	+/-	+	+	+
Shadows/shading	-	-	-	-	-	-	+/-	+/-	+	+	+
Highlights	-	-	-	-	-	-	-	-	-	-	-

colour spaces	RGB	I	Lab	rgI	HSV	rgb
Light Intensity	-	-	+/-	2/3	2/3	+
Shadows/shading	-	-	+/-	2/3	2/3	+
Highlights	-	-	-	-	1/3	-

Table 1: The invariance properties of both the individual channels and the colour spaces used in this paper, sorted by degree of invariance. A “+/-” means partial invariance. A 1/3 means that one of the three colour channels is invariant.

these images we rely on the other diversification method of using good object locations.

In this paper we always use a single colour space throughout the algorithm, meaning that both the initial grouping algorithm [13] and our subsequent grouping algorithm are performed in the same colour space.

Complementary Similarity Measures. We define four complementary, fast-to-compute similarity measures. These measures are all in the range [0, 1] which facilitates combinations of these measures.

$s_{\text{colour}}(r_i, r_j)$ measures colour similarity. Specifically, for each region we obtain one-dimensional colour histograms

Extract object location boxes L from all regions in R

For the similarity $s(r_i, r_j)$ between region r_i and r_j we want a variety of complementary measures under the constraint that they are fast to compute. In effect, this means that the similarities should be based on features that can be propagated through the hierarchy, *i.e.* when merging region r_i and r_j into r_t , the features of region r_t need to be calculated from the features of r_i and r_j without accessing the image pixels.

3.2 Diversification Strategies

The second design criterion for selective search is to diversify the sampling and create a set of complementary strategies whose locations are combined afterwards. We diversify our selective search (1) by using a variety of colour spaces with different invariance properties, (2) by using different similarity measures s_{ij} , and (3) by varying our starting regions.

Complementary Colour Spaces. We want to account for different scene and lighting conditions. Therefore we perform our hierarchical grouping algorithm in a variety of colour spaces with a range of invariance properties. Specifically, we use the following colour spaces with an increasing degree of invariance: (1) *RGB*, (2) the intensity (grey-scale image) *I*, (3) *Lab*, (4) the *rg* channels of normalized *RGB* plus intensity denoted as *rgI*, (5) *HSV*, (6) normalized *RGB* denoted as *rgb*, (7) *C* [14] which is an opponent colour space where intensity is divided out, and finally (8) the Hue channel *H* from *HSV*. The specific invariance properties are listed in Table 1.

Of course, for images that are black and white a change of colour space has little impact on the final outcome of the algorithm. For

This leads to a colour histogram $C_i = \{c_i^1, \dots, c_i^n\}$ for region r_i with dimensionality $n = 75$ when three colour channels are used. The colour histograms are normalised using the L_1 norm. Similarity is measured using the histogram intersection:

$$s_{\text{colour}}(r_i, r_j) = \sum_{k=1}^n \min(c_i^k, c_j^k).$$

The colour histograms can be efficiently propagated through the hierarchy by

$$C_t = \frac{\text{size}(r_i) \times C_i + \text{size}(r_j) \times C_j}{\text{size}(r_i) + \text{size}(r_j)}.$$

The size of a resulting region is simply the sum of its constituents: $\text{size}(r_t) = \text{size}(r_i) + \text{size}(r_j)$.

$s_{\text{texture}}(r_i, r_j)$ measures texture similarity. We represent texture using fast SIFT-like measurements as SIFT itself works for material recognition [20]. We take Gaussian derivatives in eight orientations using $\sigma = 1$ for each colour channel. For each orientation for each colour channel we extract a histogram using a bin size of 10. This leads to a texture histogram $T_i = \{t_i^1, \dots, t_i^n\}$ for each region r_i with dimensionality $n = 240$ when three colour channels are used. The histograms are normalised using the L_1 norm. Similarity is measured using histogram intersection:

$$s_{\text{texture}}(r_i, r_j) = \sum_{k=1}^n \min(t_i^k, t_j^k).$$

Texture histograms are efficiently propagated through the hierarchy in the same way as the colour histograms.

4

$s_{\text{size}}(r_i, r_j)$ encourages small regions to merge early. This forces regions in S , *i.e.* regions which have not yet been merged, to be of similar sizes throughout the algorithm. This is desirable because it ensures that object locations at all scales are created at all parts of the image. For example, it prevents a single region from gobbling up all other regions one by one, yielding all scales only at the location of this growing region and nowhere else. $s_{\text{size}}(r_i, r_j)$ is defined as the fraction of the image that r_i and r_j jointly occupy:

$$s_{\text{size}}(r_i, r_j) = 1 - \frac{\text{size}(r_i) + \text{size}(r_j)}{\text{size}(im)}, \quad (4)$$

where $\text{size}(im)$ denotes the size of the image in pixels.

$s_{\text{fill}}(r_i, r_j)$ measures how well region r_i and r_j fit into each other. The idea is to fill gaps: if r_i is contained in r_j it is logical to merge these first in order to avoid any holes. On the other hand, if r_i and r_j are hardly touching each other they will likely form a strange region and should not be merged. To keep the measure fast, we use only the size of the regions and of the containing boxes. Specifically, we define BB_{ij} to be the tight bounding box around r_i and r_j . Now $s_{\text{fill}}(r_i, r_j)$ is the fraction of the image contained in BB_{ij} which is not covered

by the object hypothesis set, depending on the computational efficiency of the subsequent feature extraction and classification methods.

We choose to order the combined object hypotheses according to the order in which the hypotheses were generated in the individual grouping strategy. However, as we combine results up to 80 different strategies, such an order would tend to favour large regions. To prevent this, we include some randomness as follows. Given a grouping strategy j , let r_i^j be the region created at position i in the hierarchy, where $i = 1$ represents the top of the hierarchy (whose corresponding region covers the complete image). We now calculate the position value v_i^j as $v_i^j = \text{RND} \times \text{size}(r_i^j)$ where RND is a random number in range $[0, 1]$. The final order is obtained by ordering the regions using v_i^j .

When we use locations in terms of bounding boxes, we use all the locations as detailed above. Only afterwards we remove lower ranked duplicates. This ensures that duplicate boxes have a better chance of obtaining a high rank. This is desirable if multiple grouping strategies suggest the same box location likely to come from a visually coherent part of the image.

4 Object Recognition using Selective Search

r_i and r_j :

$$fill(r_i, r_j) = 1 - \frac{size(BB_{ij}) - size(r_i) - size(r_j)}{size(im)} \quad (5)$$

We divide by $size(im)$ for consistency with Equation 4. Note that this measure can be efficiently calculated by keeping track of the bounding boxes around each region, as the bounding box around two regions can be easily derived from these.

In this paper, our final similarity measure is a combination of the above four:

$$s(r_i, r_j) = a_1 s_{colour}(r_i, r_j) + a_2 s_{texture}(r_i, r_j) + a_3 s_{size}(r_i, r_j) + a_4 s_{fill}(r_i, r_j), \quad (6)$$

where $a_i \in \{0, 1\}$ denotes if the similarity measure is used or not. As we aim to diversify our strategies, we do not consider any weighted similarities.

Complementary Starting Regions. A third diversification strategy is varying the complementary starting regions. To the best of our knowledge, the method of [13] is the fastest, publicly available algorithm that yields high quality starting locations. We could not find any other algorithm with similar computational efficiency so we use only this oversegmentation in this paper. But note that different starting regions are (already) obtained by varying the colour spaces, each which has different invariance properties. Additionally, we vary the threshold parameter k in [13].

3.3 Combining Locations

In this paper, we combine the object hypotheses of several variations of our hierarchical grouping algorithm. Ideally, we want to order the object hypotheses in such a way that the locations which are most likely to be an object come first. This enables one to find a good trade-off between the quality and quantity of the resulting

This paper uses the locations generated by our selective search for object recognition. This section details our framework for object recognition.

Two types of features are dominant in object recognition: histograms of oriented gradients (HOG) [8] and bag-of-words. HOG has been shown to be successful in combination with a linear model by Felzenszwalb *et al.* [12]. However, as the exhaustive search, HOG features in combination with a linear classifier is the only feasible choice from a computational perspective. In contrast, our selective search enables the use of more efficient and potentially more powerful features. Therefore we use words for object recognition [16, 17, 34]. However, we use a powerful (and expensive) implementation than [16, 17, 34] by employing a variety of colour-SIFT descriptors [32] and a fine pyramid division [18].

Specifically we sample descriptors at each pixel on a $\sigma = 1.2$ pyramid. Using software from [32], we extract SIFT [21] colour SIFTs which were found to be the most sensitive to detecting image structures. Extended Opponent SIFT [31] and SIFT [32]. We use a visual codebook of size 4,000 and a pyramid with 4 levels using a 1×1 , 2×2 , 3×3 , and 4×4 pyramid. This gives a total feature vector length of 360,000. In simplification, features of this size are already used [25, 37]. a spatial pyramid results in a coarser spatial subdivision cells which make up a HOG descriptor, our features contain information about the specific spatial layout of the object. Therefore, HOG is better suited for rigid objects and our features are better suited for deformable object types.

As classifier we employ a Support Vector Machine with a histogram intersection kernel using the Shogun Toolbox [28]. To apply the trained classifier, we use the fast, approximate classification strategy of [22], which was shown to work well for Bag-of-words in [30].

Our training procedure is illustrated in Figure 3. The initial examples consist of all ground truth object windows. For negative examples we select from all object locations g

Citations (6,914)

[References \(55\)](#)

... Real-time vehicle detection is a basis for every current intelligent transport system. Traditional vehicle detection techniques include background subtraction, edge detection, and motion tracking, yet they fall short of accuracy because of the dynamic conditions induced by occlusion, varying illumination, or high traffic density [14]. Convolutional neural networks (CNN) have completely transformed the arena of fast and accurate real-time vehicle detection [15]. ...

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... Object detection is more challenging than image classification because it involves recognising and localising objects in an image [10]. Early object detection models, such as region-based CNN (R-CNN) [11] used region proposal algorithms such as selective search [45] to propose regions of interest in an image, extract features from these regions using a convolutional neural network (CNN) [34], and classify the objects and wrap them in a bounding box. Region proposals were identified as a bottleneck, which the successors of R-CNN attempted to improve on [10,36]. ...

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... The module uses the raw features generated by the convolutional and pooling layers as input to the class-specific storage module to retrieve relevant prototypes. PBR-CNN [29] is an improved version of R-CNN [35], which first detects targets and parts, then selects representative [36] regions through re-scoring and non-parametric geometric constraints. However, its selective search algorithm generates many irrelevant candidate regions, increasing computational cost. ...

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June 2019

● Behrooz Azadi · ● Michael Haslgrübler · ● Georgios Sopidis · [...] · ● Alois Ferscha

The ubiquity of wearable sensors has contributed a lot in human activity recognition. Although activities of daily living have been studied over the past decades, there is a lack of such efforts on workers' activities in manufacturing industry. In this paper, we look at a simple yet frequent task for manufacturing, screwing, and investigate the ability to recognize this repetitive task through ... [\[Show full abstract\]](#)

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Comparing Salient Object Detection Results without Ground Truth

September 2014 · Lecture Notes in Computer Science

● Long Mai · ● Feng Liu

A wide variety of methods have been developed to approach the problem of salient object detection. The performance of these methods is often image-dependent. This paper aims to develop a method that is able to select for an input image the best salient object detection result from many results produced by different methods. This is a challenging task as different salient object detection results ... [\[Show full abstract\]](#)

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Document Image Segmentation Using K-Means Clustering Technique

January 2015

Manish T Wanjari · Vikas K Yeotikar · ● Keshao Kalaskar · [...] · ● MAHENDRA P. Dhole

Clustering technique is active research field in machine learning. In Document Image Segmentation, clustering technique is one of the most famous, simple and easy to implement technique. The K-means clustering technique is most widely used technique in the literature and many researchers compare their proposed work with the results achieved by the K-means. Clustering is the process of grouping ... [\[Show full abstract\]](#)

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